

Eye-Tracking oculomotor dysfunction in children — A telerehabilitation serious game approach

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Abstract: *Disorders of attention and oculomotor control in childhood present a significant clinical challenge, for which assessment tools are often costly or inaccessible. This paper presents an innovative, non-invasive, and engaging telerehabilitation solution designed to address this issue. An integrated system has been developed that uses a standard webcam, Python, and a serious game built in the Unity engine to monitor and stimulate visual attention in children. Validation on healthy subjects demonstrated the system's robustness and high consistency, with an optimal operational distance of 60 cm from the screen, where it achieves over 90% accuracy. The platform generates clinically interpretable visual outputs, such as heatmaps and scanpaths, for an immediate understanding of visual behaviour. The primary contribution is the demonstration that precise oculomotor metrics can be obtained without advanced hardware. This offers an accessible and scalable system for remote evaluation and cognitive training. While preliminary data are promising, further clinical validation on patients with neurovisual disorders is necessary.*

Keywords: eye tracking; Unity; visual attention; oculomotor dysfunction; children; telemedicine; fixation analysis; heatmap; scanpath; webcam-based tracking; calibration; machine learning; linear multiple regression.

1. Introduction

The role of vision in child development is central, as it provides fundamental information such as the shape, colour, size, and position of objects, guiding action and supporting learning [1]. According to Signorini and Luparia, the visual system performs three interconnected functions: seeing, looking, and understanding, which, by acting in synergy, enable the child's global development at affective, motor, linguistic, and cognitive levels [2]. Consequently, alterations in visual capacity during childhood represent a problem that should not be underestimated. The use of tools capable of objectively and precisely analysing eye movements therefore becomes crucial.

In this context, Eye tracking represents an advanced technology that allows for measuring where and for how long a person looks at a specific visual stimulus. Through the analysis of fixations, gaze points, and saccadic movements, valuable information can be obtained regarding visual attention, cognitive processing, and environmental exploration patterns.

In recent years, eye tracking has gained increasing importance in various fields due to its non-invasive nature and its high capacity to provide objective and precise data. Notably, it has been successfully employed in the diagnosis and screening of numerous medical conditions [3], in home-based rehabilitation [4], and in human-computer interaction applications [5], contributing to making interventions more accessible and personalised.

Eye tracking also proves to be a fundamental tool for analysing various cognitive functions [6] involved in daily activities. This scenario frames the growing interest in its use for the treatment of the most common neurodevelopmental disorders in developmental age. Current treatments, primarily based on pharmacological therapies, exhibit several limitations: they do not guarantee an effective response in all subjects, present numerous side effects, and do not directly address cognitive deficits. Only a

portion of children fully benefit from these medications, and in many cases, there remains a certain resistance to their use in younger children.

Given these operational constraints, the path has been opened for the experimentation of alternative interventions, including technology-mediated cognitive training, particularly through video games such as serious games. In this context, eye-tracking becomes not only a means of assessment but also an active rehabilitation tool [7].

However, given that the focus has primarily been on paediatric users, the use of a telerehabilitation protocol combined with eye-tracking technology has been deemed more appropriate than standard methods. This is due to the numerous advantages of this type of remote rehabilitation, especially when applied to a younger user. Among the most significant benefits identified in this case are: improved motivation [8], enhanced engagement and well-being [9], the ability to perform exercises in a perceived safer and more comfortable environment, greater flexibility in scheduling, and reduced costs for families [10].

Therefore, the focus of this article was to develop a prototype telerehabilitation web platform, by creating a local server, that would allow for the monitoring of visual attention via eye-tracking during gaming sessions in children affected by disorders impacting ocular capabilities.

2. Related work

Eye tracking techniques may be divided into two categories: on the one hand, those based on sensors, and on the other hand, those based on image acquisition using cameras.

Sensor-based systems, for example, those that utilise the Electrooculogram (EOG), measure the electrical potentials generated by variations in the ocular dipole with electrodes strategically placed around the orbit. This approach provides a direct measurement; however, it requires specialised hardware and meticulously implemented calibration procedures[11]. Although these systems are effective in terms of immediate detection, they exhibit limitations regarding invasiveness and adaptability to more dynamic environments.

In contrast, methods based on the acquisition of RGB data and their subsequent processing allow for the recognition of the ocular region. This approach is typically structured in two phases: firstly, the localisation of the eye; and secondly, the tracking of the gaze point.

Numerous studies have focused on pattern recognition-based solutions. For instance, Raudonis et al. [3] employed Principal Component Analysis (PCA) to reduce the dimensionality of the image, followed by the classification of the pupil position using neural networks, while also highlighting the need for specific calibration processes. Alternative methods, such as those proposed by Tang and Zhang [12] and Kuo et al. [13], have adopted predictive approaches based on grey-level information and particle filters, thereby ensuring low processing times alongside commendable accuracy.

Another significant line of research exploits the properties of the reflections on the cornea commonly referred to as Purkinje images. In these systems, the vector connecting the centre of the pupil to the reflection points is utilised to estimate the direction of gaze. Studies conducted by Yang et al. [14,15] have developed systems capable of compensating for issues arising from the use of eyeglasses or other accessories, employing grey-level difference techniques and integral projections to refine both localisation and tracking.

In parallel, other approaches concentrate on detection based on the eyes shape. These methods exploit the circularity of the iris and pupil by using techniques such as the Circular Hough Transform or Gabor filters. Such methods, as illustrated in the studies by Chen and Kubo [16], Kocejko et al. [17] and Khairosfaizal and Nor'aini [18], have proven particularly promising in accurately tracing the ocular contours, though they remain sensitive to variations in illumination and the individual characteristics of the user.

Furthermore, solutions that apply the differential effect produced by illumination have also been explored, distinguishing between the 'dark pupil' and the 'bright pupil' modes. These methodologies, as detailed in the works of Yoo et al. [19] and further refined in [20], employ infrared light sources

to enhance the contrast between the pupil and other components of the image, thereby improving accuracy even under adverse environmental conditions.

A further innovative approach is represented by three-dimensional models of the eye, as proposed by Zhu and Ji [21]. These models facilitate the direct calculation of the 3-D gaze direction by implementing dynamic mapping strategies that mitigate the impact of head movements and reduce the need for frequent recalibration, although at the expense of increased algorithmic complexity.

Finally, the adoption of hybrid techniques which concurrently integrate multiple methodologies, has enabled the exploitation of the specific strengths inherent in each approach, thus achieving a balance between robustness and computational efficiency. For example, Huang et al. [22] combined criteria of intensity, shape, and size using Support Vector Machine classifiers, while Li and Wee [23] and Coetzer and Hancke [24] have developed systems that integrate the detection of corneal reflections with iterative curve-fitting algorithms, obtaining promising results for applications such as driver fatigue monitoring.

Collectively, these studies provide a solid theoretical and practical foundation for the development of increasingly sophisticated, responsive, and adaptable human-machine interfaces, thereby stimulating further research aimed at enhancing the efficiency, reliability, and user-friendliness of real-time eye tracking systems.

3. Materials and methods

3.1. Game design and mechanics

The game 'Eye Catch You!', developed in the Unity 6 environment, is designed with a horizontal scrolling layout. This was chosen to leverage the greater accuracy of the eye-tracking script in that direction, as determined during the testing phase. To enhance child engagement and reduce the drop-out rate, a marine setting was chosen, featuring various types of fish with stimulating colours [25]. These appear on the screen at random, varying in both their colour and direction of travel. Specifically, the player is tasked with capturing only the red fish, with the aim of achieving the highest possible score and rank.

The game is structured across three consecutive levels of increasing difficulty. A new type of fish to be avoided is introduced in each level, and the scrolling speed is increased only in the final level. A dynamic scoring system has been implemented, which is incremented by capturing the target red fish. Furthermore, an error count (malus) is recorded, which is visible only on the clinician's private dashboard.

On the game screen, in addition to the score, an EXIT button is always visible and can be used to end the session prematurely.

A total duration of 80 seconds was deemed appropriate, comprising 20 seconds per level and approximately 15 seconds for the initial instructions and transitions between scenes.

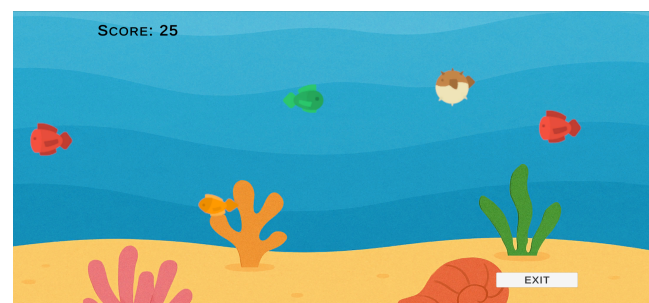


Figure 1. Unity game frame (level 3).

3.2. Server architecture

The local server on which the Reyebilitation platform is based has been developed in Python using its `http.server` module. It is configured to serve static files from specific directories, such as `"static/"`, where the HTML pages used (created with the Bootstrap library) are stored, and `"static/unity/Build/"`,

which contains the necessary files for launching the browser game developed in Unity. Specifically, the server can manage both the interface for medical professionals and one for patients, to which users are redirected upon completing the login on the game-themed customized homepage. Additionally, while the section dedicated to doctors is designed in the form of tables that allow them to consult patients under treatment, gaming sessions, and relevant details, the children's user interface focuses on illustrating the game and its narrative context, as well as providing access to the actual gameplay experience. Finally, all the data processed and transmitted to the appropriate HTML pages by the server is previously collected in an SQL database, where four pediatric patients with different visual attention or perception disorders have been initialized. Image (Fig. 2) provides a concise illustration of the server's workflow.

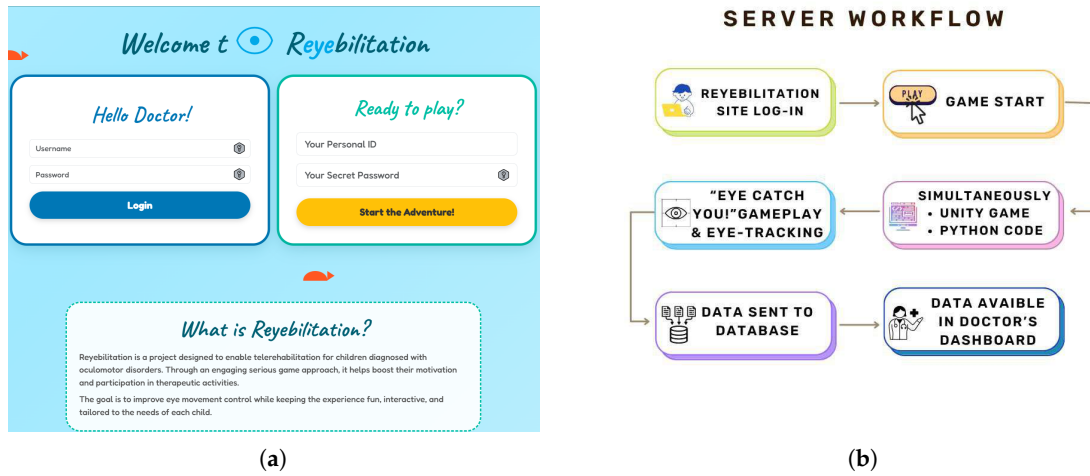


Figure 2. (a) Reyebilitation homepage, (b) Server workflow.

3.3. Eye-Tracking algorithm

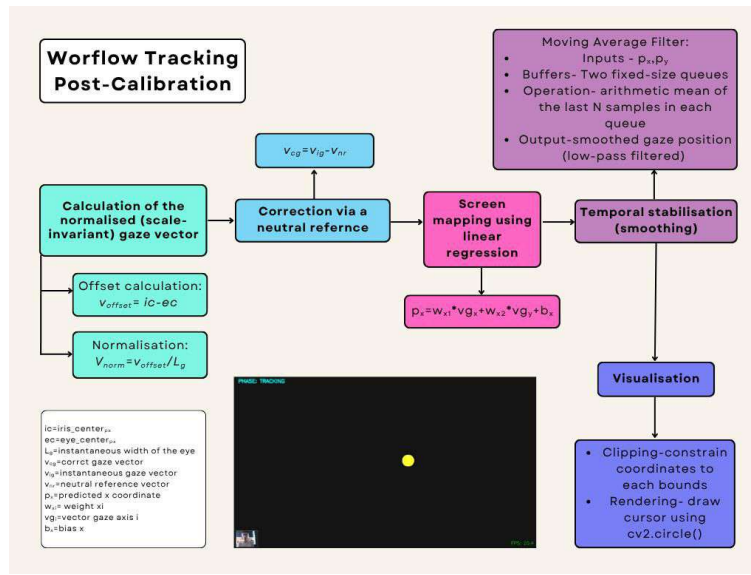
The algorithm utilizes a standard webcam and Google's MediaPipe Face Mesh library to detect eye landmarks in real-time 33 and 133 for the left eye, and 263 and 362 for the right eye. It also tracks the irises, identifying five key points per eye: one central point and four along the perimeter. Additionally, OpenCV, NumPy, and scikit-learn are employed for video processing, numerical computation, and training the calibration model using linear regression. The calibration process is structured in two phases [26]:

- **Neutral calibration (zero point):** The user focuses on a central target for approximately 50 frames, during which raw gaze vectors are collected. By calculating the average, a neutral reference vector is obtained, which is then subtracted from the instantaneous vectors to correct them:

$$g\vec{a}ze_{correct} = g\vec{a}ze_{instantaneous} - g\vec{a}ze_{neutral} \quad (1)$$

- **Screen calibration:** The user focuses on a 3x3 grid (9 points), and with each confirmation click, the system records a pair (corrected gaze vector, target coordinates). This supervised dataset is used to train two independent linear regression models, which respectively map the X and Y components of the gaze vector onto the screen. [27]

$$component_{predicted} = w_{x1}g_x + w_{x2}g_y + b_x \quad (2)$$



(a)

Figure 3. Eye Tracking workflow.

3.3.1. Data acquisition and fixations identification

During tracking, for each frame, the (x, y) coordinates of the gaze point are recorded along with the timestamp, updating a matrix that feeds into the heatmap.

At the end of the recording, the I-DT (Dispersion Threshold Identification) algorithm analyzes the spatial dispersion of the points: if the dispersion remains below a threshold for a minimum time interval, the interval is identified as a fixation event.

3.3.2. Outputs

The main outputs include statistics on fixations and gaze points. Gaze points are the individual temporal samples that indicate where the user is looking, while fixations (typically lasting between 100 and 600 ms) represent clusters of stable gaze points. The following fixation metrics are calculated [28]:

- Mean duration and standard deviation
- Minimum/maximum values
- Total duration
- Spatial stability

In addition, a quadrant analysis is conducted using a 3×3 grid, which reports the distribution, counts, and percentages of gaze points and fixations. Visualisations are also produced, including heatmaps (obtained by convolving the matrix with a Gaussian kernel), scanpaths (lines connecting fixations represented by circles whose size indicates duration and chronological order), and histograms displaying the distributions of the metrics.

Starting from the pixel coordinates (x, y) of the raw gaze for both the left and right eye, we derive the gaze angle (theta), defined as the angle of view of a point on the screen relative to the user's eyes. It is calculated as follows [29]:

$$\text{Gaze angle } (\theta) = \arctan\left(\frac{\text{OSD}}{Z}\right) \quad (3)$$

Here Z is the distance from the eye to the screen and OSD represents the distance from the centre of the screen.

3.4. Methods validation

The validation was carried out under controlled conditions using static tests.

- **Accuracy test:** for N=9 fixed targets, for each fixation (confirmed by a click) the mean position is calculated over the k immediately preceding samples. The accuracy error is defined as the

Euclidean distance between the target centre and this mean position, with its mean and standard deviation computed both in pixels and in degrees [27].

$$\bar{P}_{E,i} = \frac{1}{k} \sum_{j=t-k+1}^t P_E(j) \quad (4)$$

$$\varepsilon_i = \|P_{T,i} - \bar{P}_{E,i}\|_2 = \sqrt{(x_{T,i} - \bar{x}_{E,i})^2 + (y_{T,i} - \bar{y}_{E,i})^2} \quad (5)$$

- **Precision test:** for M=5 targets, while maintaining fixation for a time T, L consecutive samples are recorded. The precision is evaluated using the Root Mean Square (RMS) of the displacements between consecutive samples, averaged over all targets and accompanied by the corresponding standard deviation [27].

$$\text{RMS}_j = \sqrt{\frac{1}{L-1} \sum_{k=1}^{L-1} \|P_{E,i}(t_{k+1}) - P_{E,i}(t_k)\|_2^2} \quad (6)$$

4. Results & discussion

4.1. Quantitative analysis of validation

The comparative analysis of the validation tests (Table 1), conducted at four different user–screen distances (30, 45, 60, and 75 cm), reveals that performance depends on this parameter. The system’s accuracy at proximal distances (30 cm) shows an increase in angular error, attributable to the wide eye rotations required to cover the entire screen, which makes it difficult to approximate a linear regression model. Conversely, at greater distances (75 cm), the decline in performance is due to the reduced spatial resolution of the face and irises in the frame captured by the webcam, compromising the landmark detection and consequently the validity of the calibration model, as evidenced by the drastic increase in pixel error. Measurement uncertainty regarding accuracy reaches its minimum at 60 cm ($\pm 1^\circ$), indicating that at this distance the system is more accurate, reliable, and uniform in its performance. As for precision, a notable stability in angular error is observed, remaining consistently around 1° across all tested distances. This suggests that the intrinsic frame-to-frame algorithm noise is less sensitive to the operational distance than the accuracy of the overall mapping.

Static accuracy test results

Distance (cm)	Mean Accuracy (%)	Mean Error (px) $\pm$ std	CI 95% (px)	Mean Error (deg) $\pm$ std	CI 95% (deg)
30	94%	83 $\pm$ 46	53 – 113	4 $\pm$ 2	3 – 6
45	94%	84 $\pm$ 39	59 – 109	3 $\pm$ 1	2 – 4
60	95%	70 $\pm$ 29	51 – 89	2 $\pm$ 1	1 – 2
75	90%	144 $\pm$ 70	98 – 190	3 $\pm$ 1	2 – 4

Static precision test results

Distance (cm)	Mean Precision (%)	Average RMS Jitter (px) $\pm$ std	Average RMS Jitter (deg) $\pm$ std
30	98%	26 $\pm$ 5	1 $\pm$ 0
45	98%	34 $\pm$ 26	1 $\pm$ 1
60	97%	44 $\pm$ 19	1 $\pm$ 1
75	98%	25 $\pm$ 3	1 $\pm$ 0

Table 1. Accuracy and precision test results for User 1.

The acquisition of a second validation dataset, related to a different user (Table 2), confirms and reinforces the preliminary observations by consolidating the hypothesis of an optimal operational window for the system. For this user as well, a minimum of $2^\circ \pm 1^\circ$ is achieved at 60 cm. Although the

average angular error at 75 cm ($2^\circ \pm 1^\circ$) appears comparable to that at 60 cm, the pixel error analysis at 75 cm versus 60 cm reveals a significant degradation in performance. This phenomenon confirms that at greater distances even a seemingly modest angular error translates into a much larger and less reliable spatial inaccuracy on the screen. Moreover, the uncertainty in accuracy reaches its absolute minimum at 60 cm, both in pixels (± 24 px) and in degrees ($\pm 1^\circ$), further validating this distance as the operational "sweet spot" where the system exhibits maximum consistency and reliability.

Static accuracy test results

Distance (cm)	Mean Accuracy (%)	Mean Error (px) $\pm$ std	CI 95% (px)	Mean Error (deg) $\pm$ std	CI 95% (deg)
30	94%	91 $\pm$ 39	66 – 117	5 $\pm$ 2	3 – 6
45	92%	116 $\pm$ 53	81 – 150	4 $\pm$ 2	3 – 5
60	95%	79 $\pm$ 24	64 – 95	2 $\pm$ 1	2 – 2
75	93%	110 $\pm$ 55	74 – 145	2 $\pm$ 1	2 – 3

Static precision test results

Distance (cm)	Mean Precision (%)	Average RMS Jitter (px) $\pm$ std	Average RMS Jitter (deg) $\pm$ std
30	98%	26 $\pm$ 8	1 $\pm$ 0
45	96%	51 $\pm$ 7	2 $\pm$ 0
60	97%	51 $\pm$ 4	1 $\pm$ 0
75	96%	58 $\pm$ 12	1 $\pm$ 0

Table 2. Accuracy and precision test results for User 2.

The consistency between users and the replicability of the results with respect to distance indicate that the observed limitations are intrinsic characteristics of the system, linked to the scene geometry, sensor, and calibration model, rather than mere random artefacts. Finally, thanks to the mapping model based on multiple linear regression applied to the 9-point calibration, an average accuracy and precision of over 90% is achieved regardless of the subject's distance from the camera.

4.2. Qualitative analysis of clinical parameters

Warm colours (yellow) indicate areas of greatest visual attention, that is, points with the maximum number or longest duration of fixations, while cool colours (violet/blue) correspond to a minimal level of attention. This representation is useful for rapidly identifying visual preferences, neglected areas, or determining whether attention is focused on a specific point or dispersed across multiple elements of the stimulus.

The scanpath helps analyze a user's visual behavior by tracking saccade directions to understand attention flow and revisited areas. Overlaying it onto a key video game frame reveals whether relevant elements were noticed or overlooked [30].

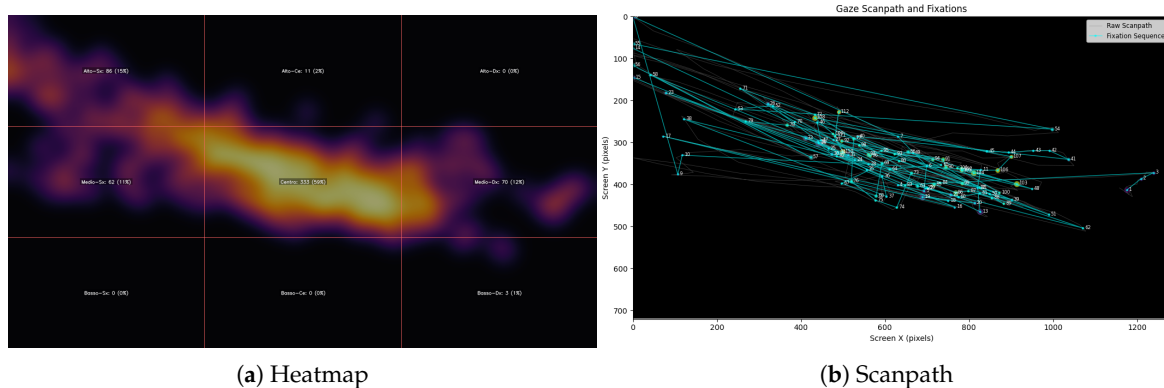


Figure 4. Demonstration results obtained from a healthy adult individual.

Histograms provide a more informative visual representation than mere numerical values, allowing for the rapid identification of error concentrations, outliers, and any irregular distributions. They are particularly useful for comparing the similarity between data collected from different eye trackers.

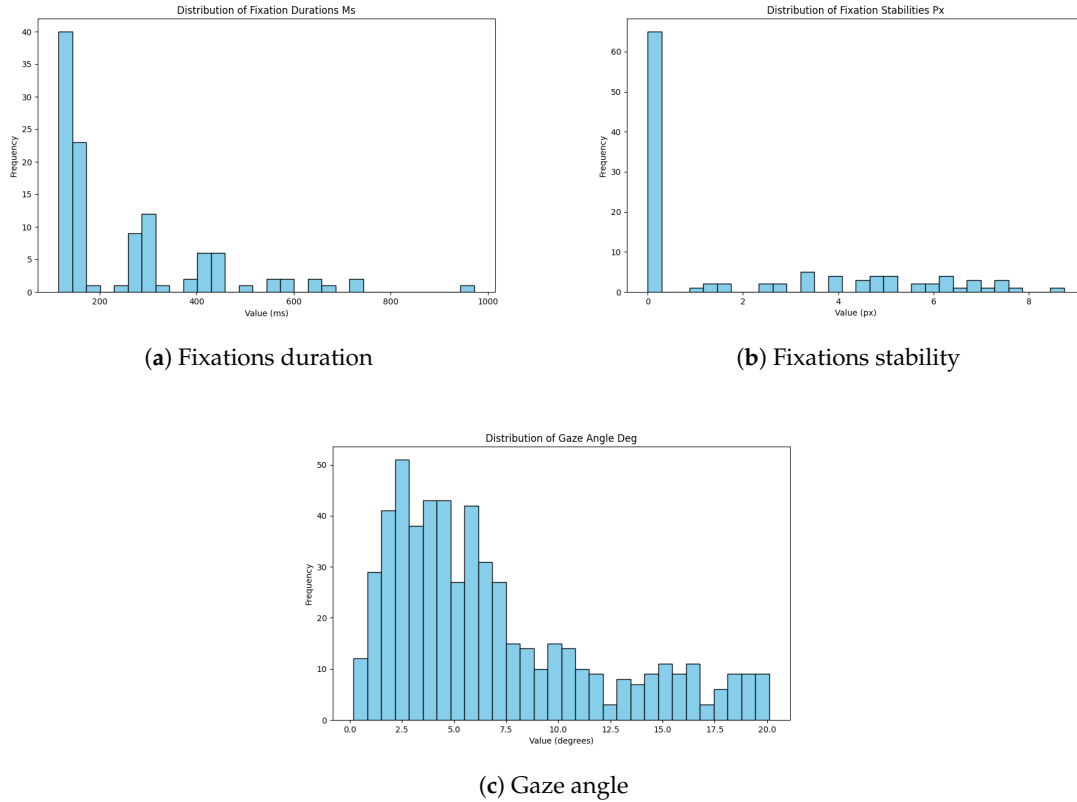


Figure 5. Histograms with the results obtained from a healthy adult individual.

A textual report is provided to clinicians, detailing parameters such as gaze point count low values may indicate tracking issues or poor compliance. Fixation duration reflects attention: very short durations suggest rapid, disorganized exploration, while long ones indicate deep processing or difficulty shifting focus. High deviation in fixation durations could signal fluctuating attention. Fixation stability assesses oculomotor control, with low values hinting at difficulty maintaining precise focus. Quadrant analysis identifies focus patterns, highlighting areas the subject may ignore. A high average gaze angle suggests frequent deviations from the central line of sight.

4.3. Principal contribution and future developments

The project demonstrates that it is possible to obtain precise and reliable oculomotor metrics using a simple webcam and a video game developed in Unity, without the need for expensive tools such as depth cameras. Thanks to a nine-point calibration and a multiple linear regression model, the system provides visual and textual outputs that are easy to interpret clinically, thereby supporting telemedicine and cognitive rehabilitation. The promising results have, however, been limited by the scarcity of subjects with specific disorders and reduced computational resources. Future developments aim at clinical validation on real patients, the use of advanced machine learning algorithms, and system optimisation for real-time applications.

5. Conclusions

Attention and oculomotor control disorders in childhood represent a significant clinical challenge, often addressed with costly or inaccessible tools. This study proposes an innovative, non-invasive, accessible, and engaging solution by developing an integrated system between Unity and Python

for eye tracking using a standard webcam, aimed at monitoring and stimulating visual attention in children through a video game. The primary scientific contribution lies in demonstrating that, even without advanced hardware, reliable measurements of fixations, saccades, and visual distributions can be obtained through nine-point calibration and multiple linear regression models. The metrics collected, validated on healthy subjects, show high user consistency and good precision at specific operational distances (optimal at 60 cm), supporting the system's robustness. The visual representations obtained (heatmaps, scanpaths, histograms) allow for an immediate clinical interpretation of attention patterns. Although these data were collected from healthy subjects, they suggest the tool's potential usefulness in cognitive rehabilitation and telemedicine. While preliminary data are promising, clinical validation on patients with neurovisual disorders will be necessary, along with the integration of machine learning and deep learning models and the enhancement of computational infrastructure. This will support more complex and accurate real-time analyses, making the system more effective and scalable, useful for both evaluation and remote cognitive training.

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